PROJECT ALTAIR / Software Recruitment

Phase 3 Assessment: The Turtle Has Field Duty (ROS 2)

Project Altair Software Subteam
Recruitment Batch 24 · October 2026

CANDIDATE NAME	STUDENT ID	DEPARTMENT	TARGET ROS 2 ENV
Mihiya	240041128	CSE	Ubuntu 22.04 (Humble)

Primary Submission Portals (Clickable)

Phase 3 — GitHub Repository

- Package: altair_phase3 (ament_python)
- Complete sources, metadata & launch file
- Automated controller & state test suite

Open GitHub Repository ↗

Phase 3 — Google Drive Folder

- Screen recordings with voice narration
- Tasks 1, 2a, 2b, 3, and Task 4 Reel
- Architectural diagrams & terminal logs

Open Google Drive Folder ↗

Task Map & Implemented Deliverables (100 Marks)

Task	Marks	Node(s) & Topic / Interface	Assessed Technical Behaviour & Evidence

1. Rover Telemetry	25	`mars_telemetry_publisher`
		`mission_control_monitor`
		Topic: /rover/telemetry (String)
			• 1 Hz JSON telemetry: battery %, temp °C, speed m/s, elapsed s.
			• Automatic prominent warning triggered when battery < 25%.
			• Demonstrates parameter modification & message journey.
			• Video duration: 60-90s with voice explanation.

2a. Turtle Driving Menu	10	`turtle_driver` Topic: /turtle1/cmd_vel Type: geometry_msgs/Twist	• Non-blocking menu: P (Patrol), R (Rectangle), Z (Zigzag), Q (Quit). • Smooth timed movements; clean stop between patterns. • Robust: invalid characters handled cleanly without crashing.
2b. Stop Not Sticking	15	**Architectural Analysis** `CLI: ros2 topic info --verbose` Design: twist_mux & Schmitt Hysteresis Supervisor	• Root cause: 10 Hz nav messages overwrite single zero stop pulse. • CLI proves endpoint existence (pub count=2), but not rates/order. • Robust command ownership: supervisor mux with latched pause. • Dual-threshold distance buffer prevents threshold flickering.

3. Three Stops Before Cha	35	`field_mission`
		`field.launch.py`
		Topic: /field_blocked (Bool)
		Srv: /resume_mission (Trigger)
		Pose: /turtle1/pose
			• Closed-loop waypoint navigation: Bottle (8.0, 5.5) -> Mallet (8.0, 8.0) -> Base (5.5, 5.5). Arrival tolerance <= 0.3 units.
			• Mid-leg obstacle: halts immediately on /field_blocked=True.
			• NO auto resume: clearing flag holds pause; rejects resume while blocked.
			• Explicit resume via /resume_mission completes remaining targets.

4. Explain Like a Reel	15	**Video Concept Reel (60-90s)** Analogy: FM Radio Broadcast Live parameter change & demo	• Clear pub-sub explanation using local radio station analogy. • Demonstrates real nodes & explains multi-subscriber decoupling. • Highlights initial misconception (queue vs volatile stream).

Phase 2 — Previously Missed / Prior Work Reference

- Candidate Student ID: 240041128 I Batch 24 Assessment Score: 9.00 / 9.00 on personal map.
- Completed Sections: S01 A* Planning & EKF, S02 Murphy's Law Autonomy, S03 Radio Packet Loss, S04 Perception.
- All Phase 2 solution scripts, notebooks, and verification videos are preserved in the shared Drive folder.
- **Drive Link: https://drive.google.com/drive/folders/1kQfYiVkHJaHs8_0cvoqsOJe7xtQ1a612**